

all-sites-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(scales)
library(readxl)
library(patchwork)
library(ggbridges)
library(lme4)
library(lmerTest)
library(knitr)
library(rmarkdown)

library(ragg)
```

Data

```
# Trail Creek Isotope
compiled_isotope <- read_csv(here("data",
"compiled_river_reach_isotope_data.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_data <- read_csv(here("data", "tc_gradient_data.csv"))

trail_creek <- compiled_isotope |>
  filter(location %in% c("utc", "tc"))
```

```
coal_creek <- compiled_isotope |>
  filter(location == "cc")

italian_creek <- compiled_isotope |>
  filter(location == "ic")
```

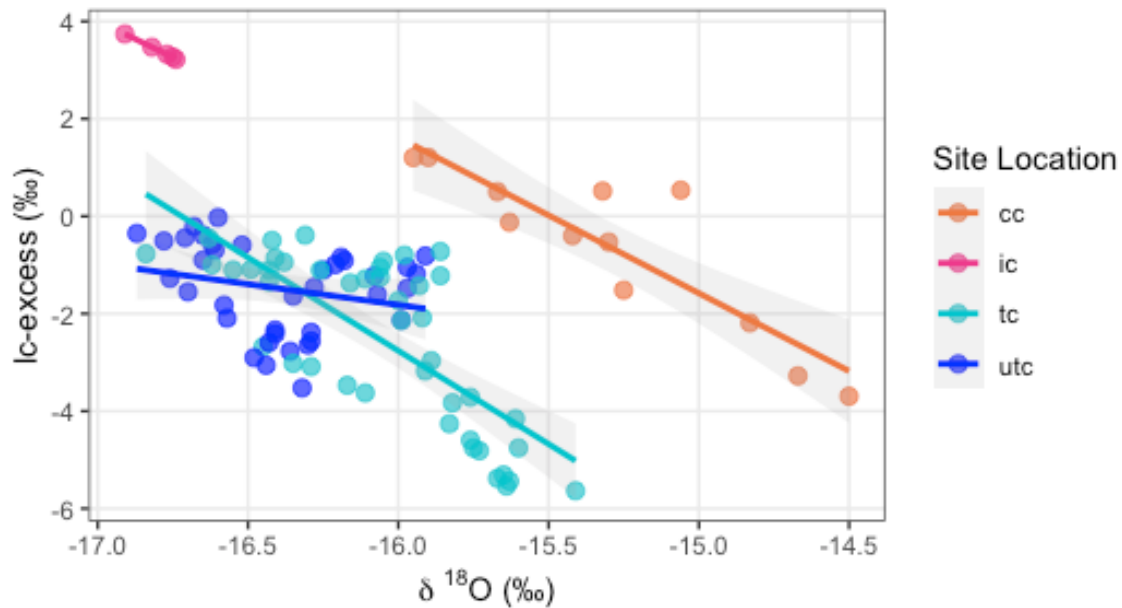
Exploratory Visuals with ALL Sites

Line-Conditioned Excess (lc-excess) vs Delta 18O

```
ggplot(data = compiled_isotope, aes(x = delta_o, y = lc_excess, color = location))
+
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  scale_color_manual(values = c("tc" = "turquoise3", "cc" = "sienna2", "ic" =
"violetred2", "utc" = "blue")) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " ^18, "0 (\u2030)")), # Formats as  $\delta^{18}O$  (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
    legend.position = "right",
    panel.grid.minor = element_blank())
```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals between sites



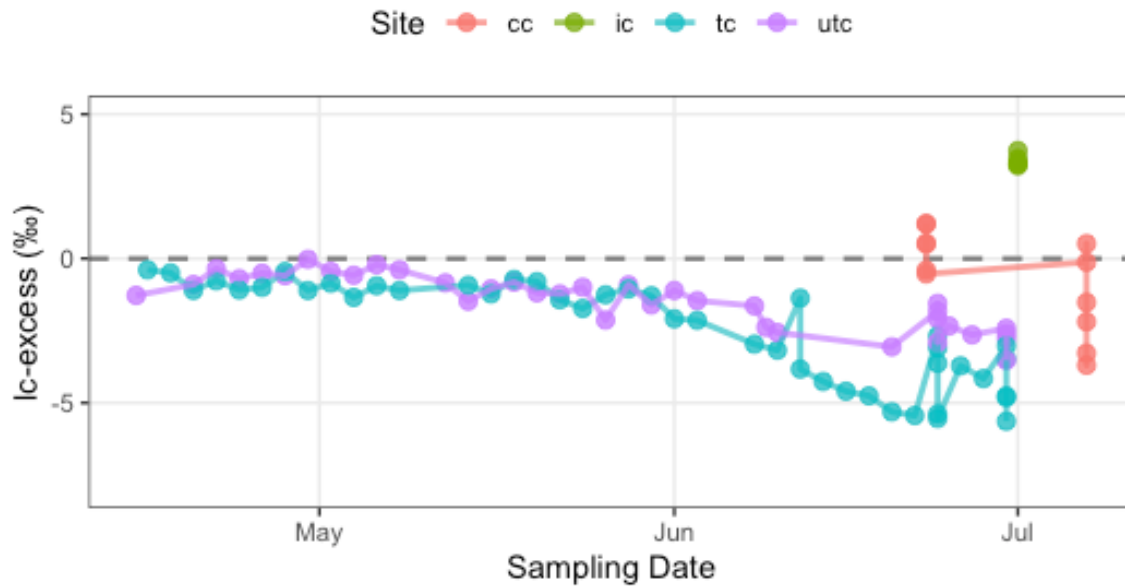
Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = compiled_isotope,
  aes(x = date,
    y = lc_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(linewidth = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",
    y = "lc-excess (‰)",
    color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
```

```
panel.grid.minor = element_blank(),
legend.position = "top")
```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment



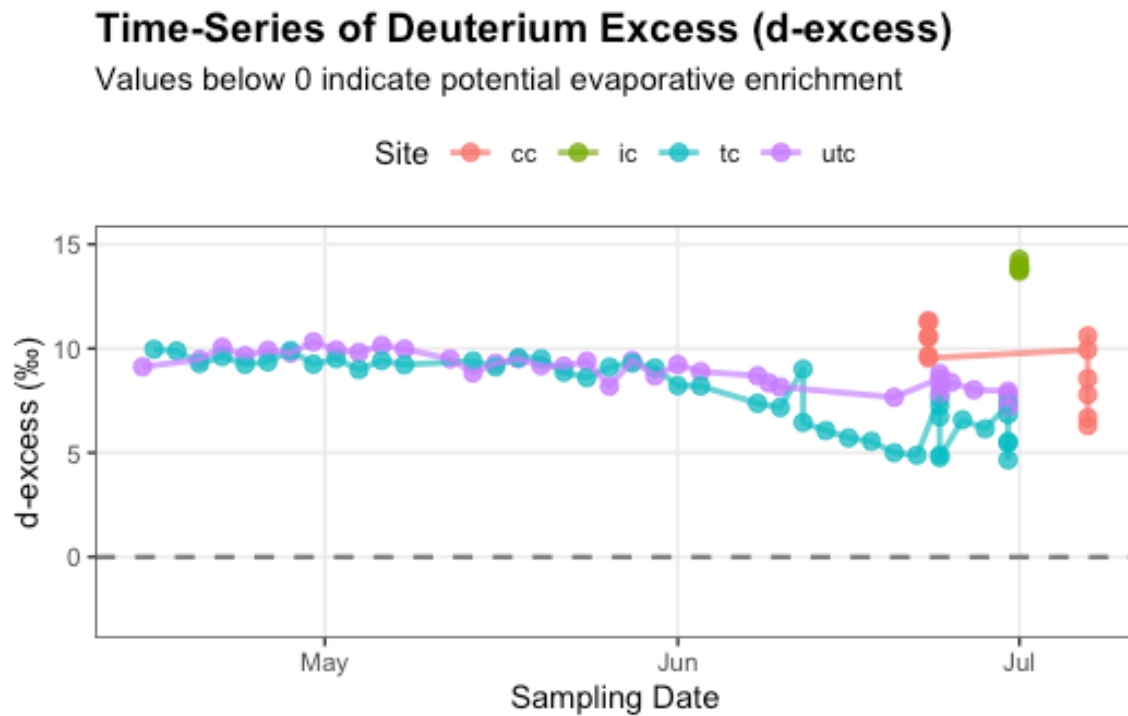
Time-Series of Deuterium Excess (d-excess)

```
ggplot(data = compiled_isotope,
  aes(x = date,
    y = d_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(size = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Deuterium Excess (d-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",
    y = "d-excess (‰)",
```

```

    color = "Site") +
  ylim(-3,15) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")

```



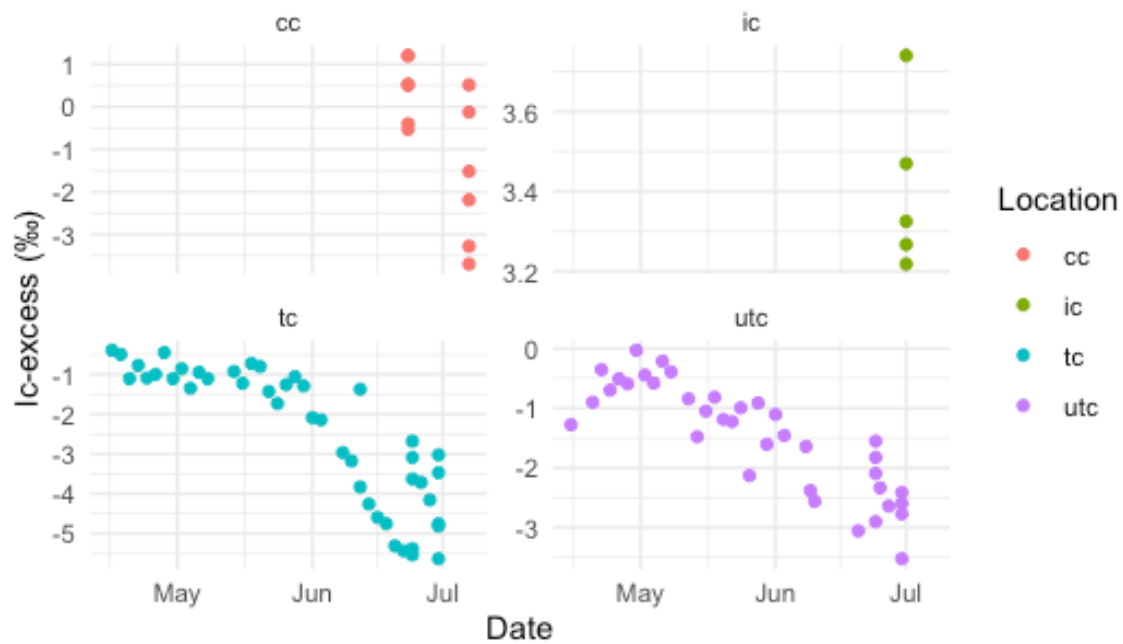
Temporal Variations in Line-Conditioned Excess (by location)

```

ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = location)) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (%)",
    color = "Location"
  ) +
  facet_wrap(~location, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))

```

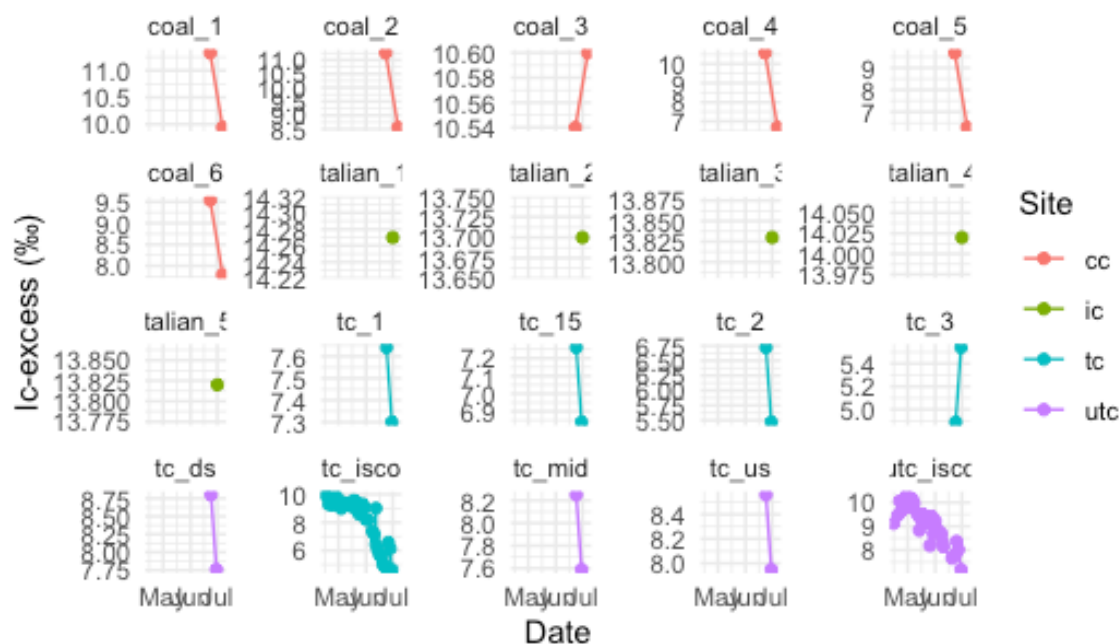
Temporal Variations in Line-Conditioned Excess



Temporal Variations in Deuterium Excess (by sites)

```
ggplot(compiled_isotope, aes(x = date, y = d_excess, color = location)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Deuterium Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Temporal Variations in Deuterium Excess

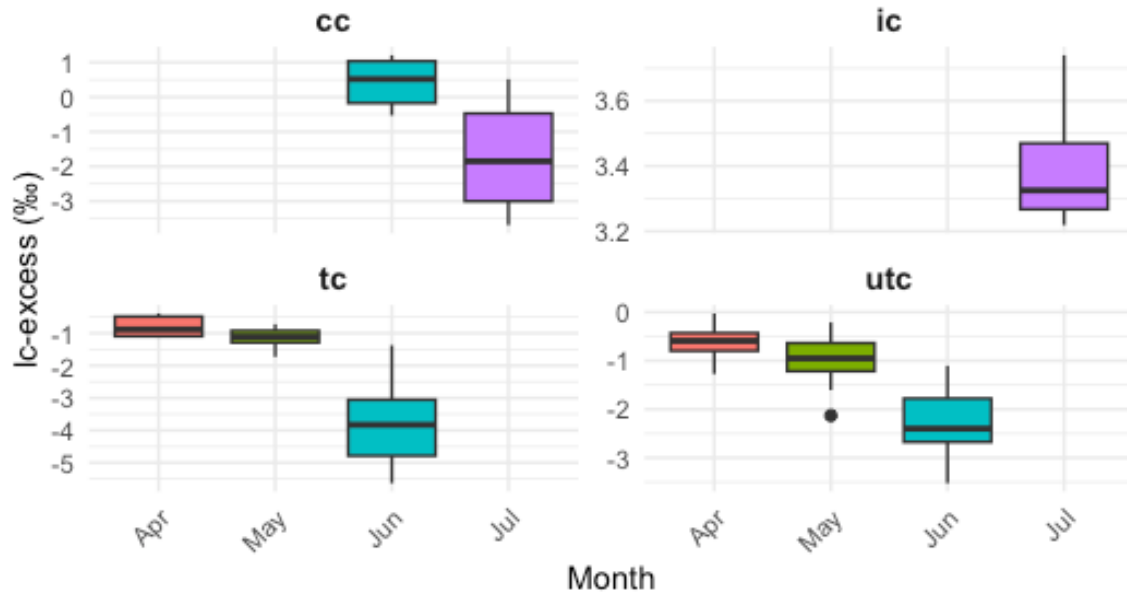


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(compiled_isotope, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (%)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1)) # tilting the x axis text
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



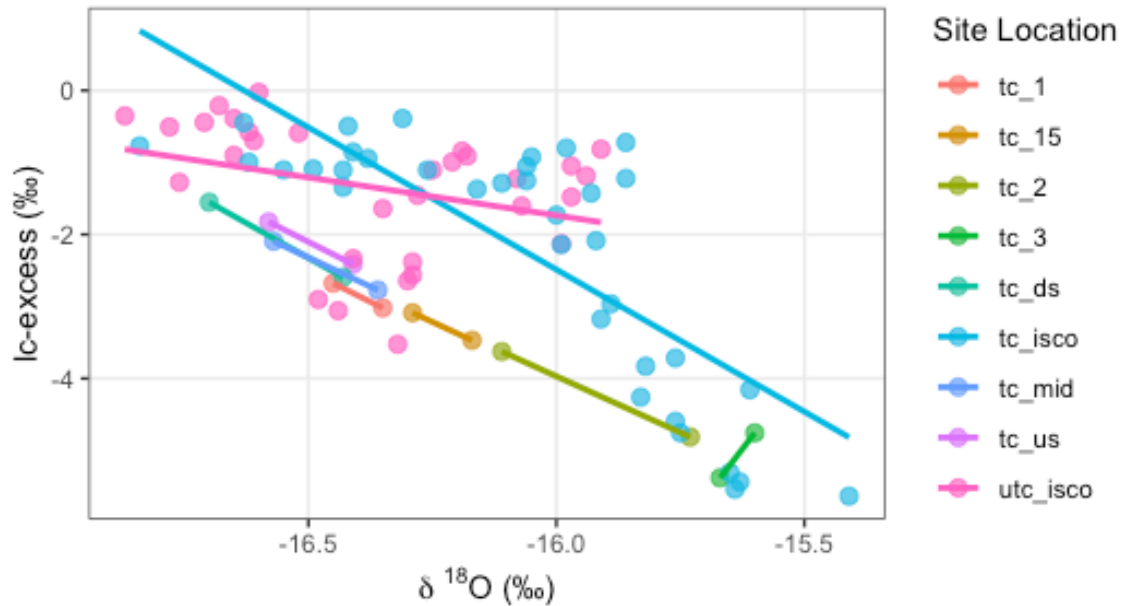
Exploratory Visuals of Trail Creek

Line-Conditioned Excess (lc-excess) vs. Delta 18O

```
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, alpha = 0.15, linewidth = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " ^18, " "0 (\u2030)")), # Formats as  $\delta^{18}O$  (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())
```

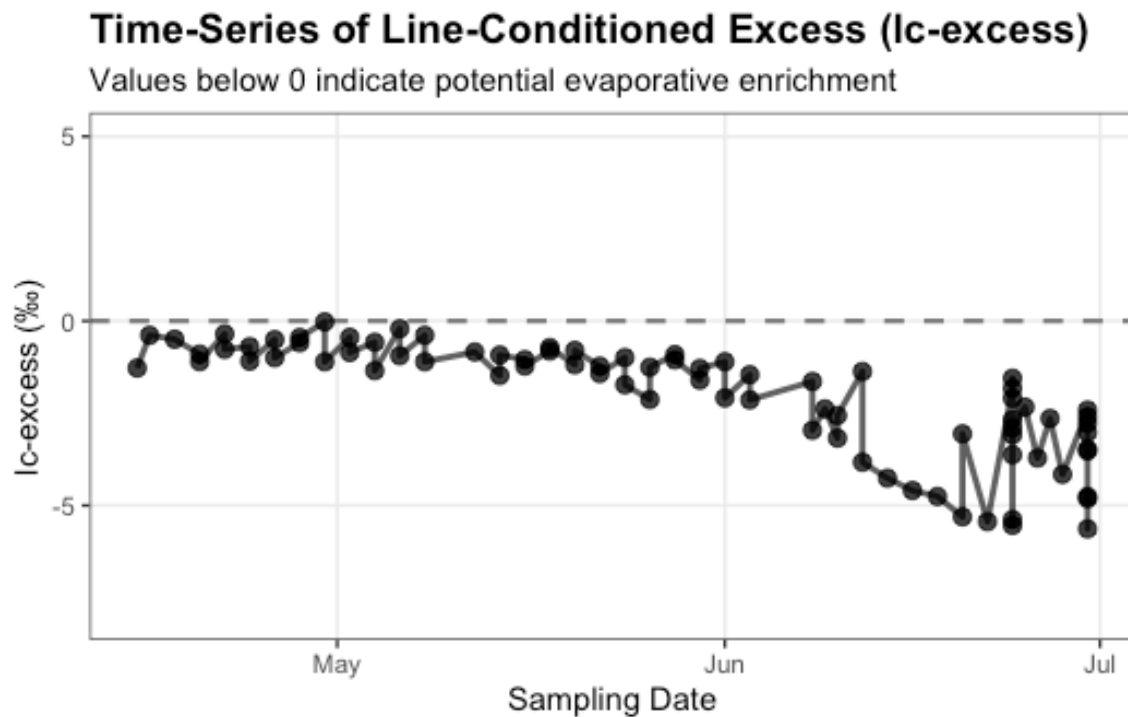

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals between sites



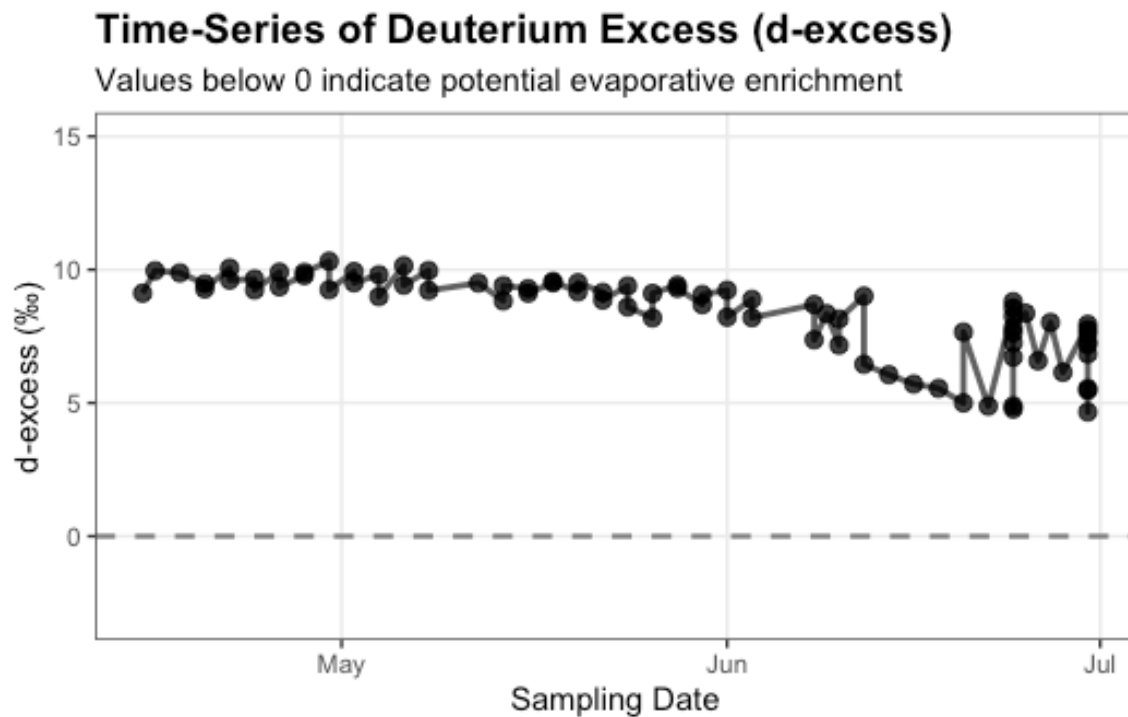
Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = trail_creek,
       aes(x = date,
           y = lc_excess)) +
  geom_hline(yintercept = 0,
            linetype = "dashed",
            color = "gray50",
            size = 0.8) +
  geom_point(size = 2.5,
            alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (‰)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



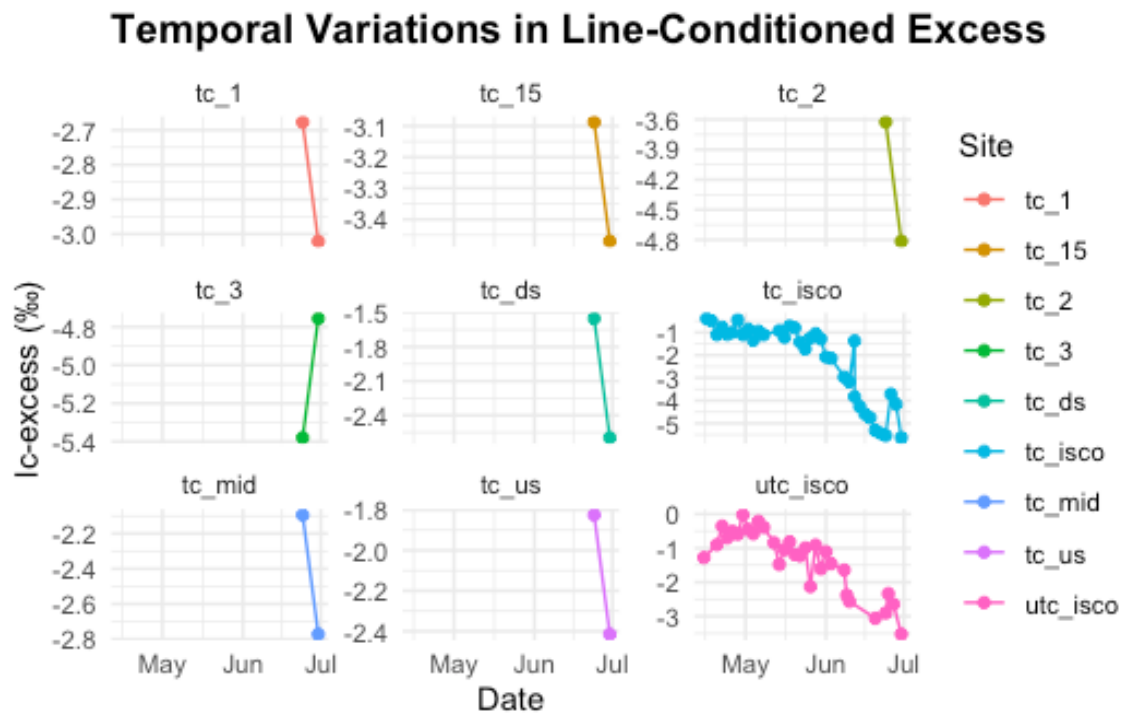
Time-Series of Deuterium Excess (d-excess)

```
ggplot(data = trail_creek,
  aes(x = date,
    y = d_excess)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(size = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Deuterium Excess (d-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
    x = "Sampling Date",
    y = "d-excess (‰)",
    color = "Site") +
  ylim(-3,15) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    legend.position = "top")
```



Temporal Variations in Line-Conditioned Excess

```
ggplot(trail_creek, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (%)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

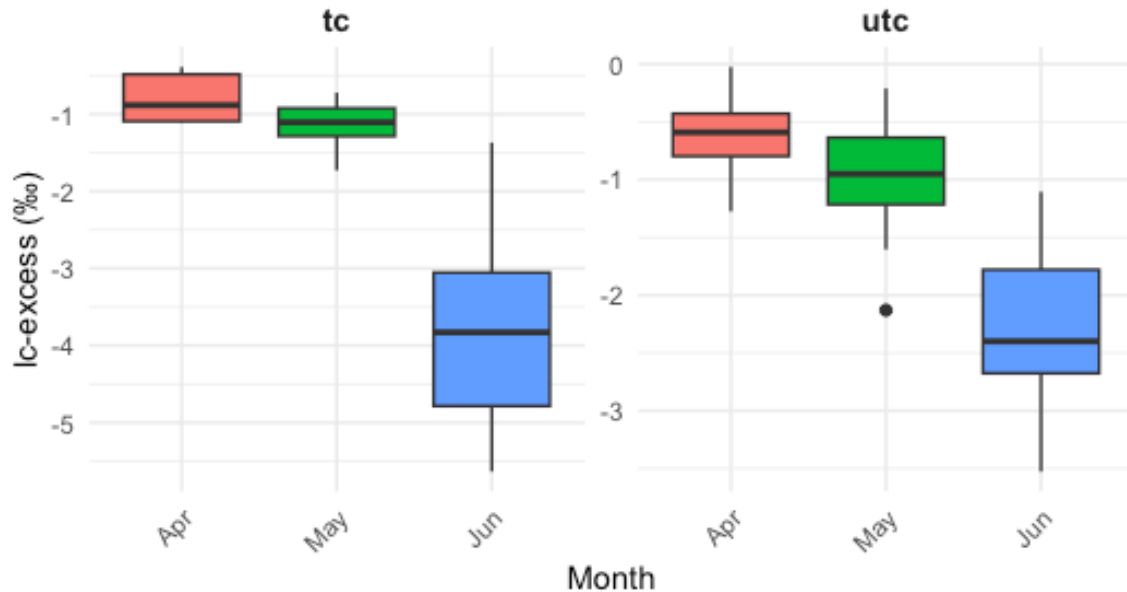


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(trail_creek, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y =
lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11), # Emphasize site names
    axis.text.x = element_text(angle = 45, hjust = 1) # Tilt month labels if
they overlap
  )
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



Dual-isotope plot ($\delta^{18}\text{O}$ vs δD) with LMWL reference

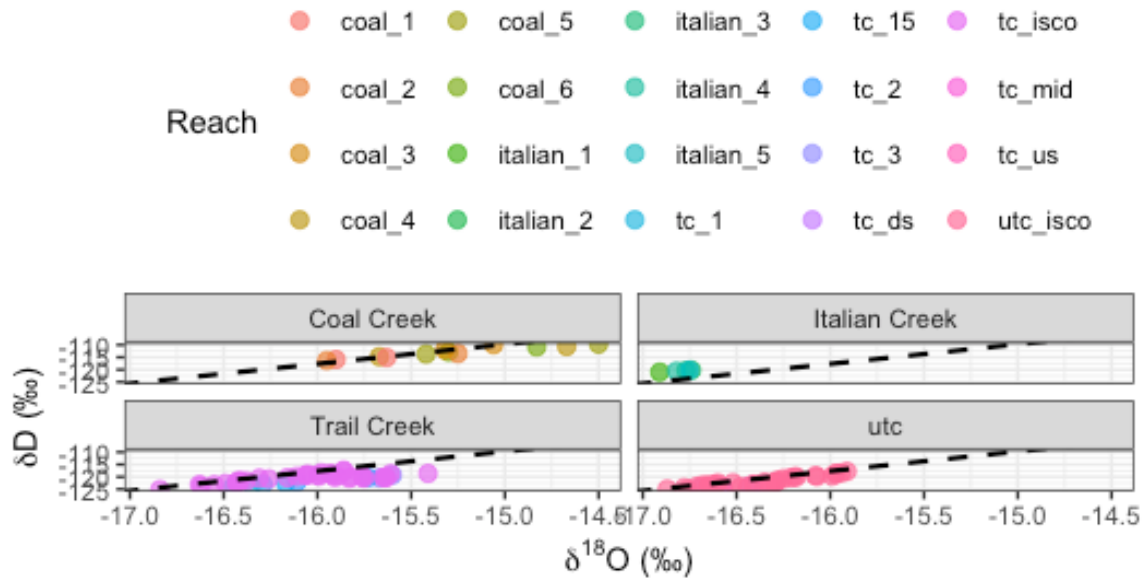
```
# Trail Creek LMWL calculated from OPIC

lmwl_slope <- 7.94
lmwl_intercept <- 9.37

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_abline(slope = lmwl_slope, intercept = lmwl_intercept,
             linetype = "dashed", color = "black", linewidth = 0.8) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         ic = "Italian
Creek")))) +
  labs(title = "Dual Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
       subtitle = "Dashed line = Trail Creek LMWL",
       x = expression(paste(delta^18, "O (\u2030)")),
       y = expression(paste(delta, "D (\u2030)")),
       color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top")
```

Dual Isotope Plot: $\delta^{18}\text{O}$ vs δD

Dashed line = Trail Creek LMWL

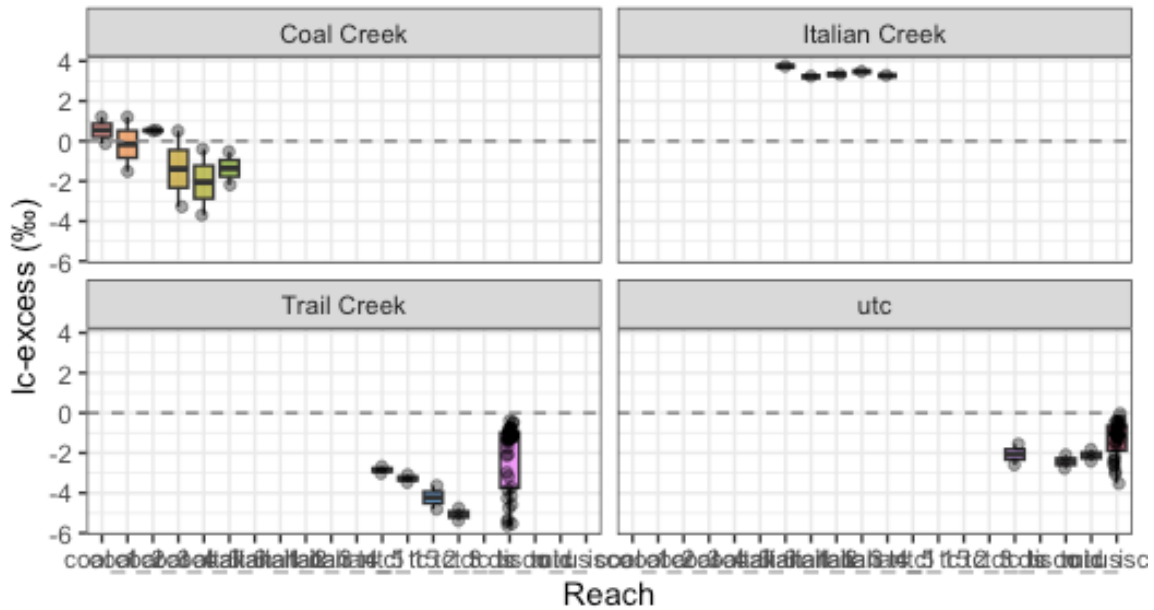


Lc-excess boxplot by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = site, y = lc_excess, fill = site)) +
  geom_boxplot(alpha = 0.7, outlier.shape = 21) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         ic = "Italian
Creek")))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  labs(title = "lc-excess by Reach Position",
       subtitle = "Values below 0 indicate evaporative enrichment",
       x = "Reach", y = "lc-excess (\u2030)") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
       legend.position = "none")
```

lc-excess by Reach Position

Values below 0 indicate evaporative enrichment

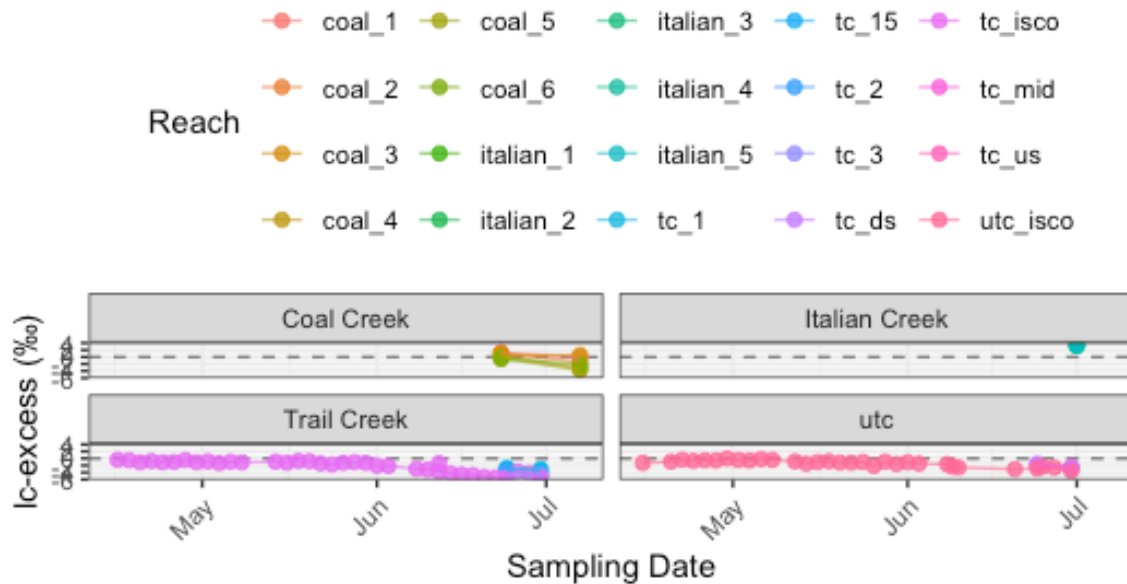


Time series of lc-excess across season, colored by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                    cc = "Coal Creek",
                                                    ic = "Italian
Creek")))) +
  labs(title = "Seasonal Trends in lc-excess",
        subtitle = "2025 low-snowpack year",
        x = "Sampling Date", y = "lc-excess (‰)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top",
        axis.text.x = element_text(angle = 45, hjust = 1))
```

Seasonal Trends in lc-excess

2025 low-snowpack year



Mixed model: $lc_excess \sim site * location$, with sample as random effect if repeated

```
# Mixed model
model_mixed <- lmer(lc_excess ~ site * location + (1 | date), data =
compiled_isotope)
summary(model_mixed)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: $lc_excess \sim site * location + (1 | date)$

Data: compiled_isotope

REML criterion at convergence: 257.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.97083	-0.32109	0.04244	0.42476	1.72105

Random effects:

Groups	Name	Variance	Std.Dev.
date	(Intercept)	1.6758	1.2945
Residual		0.4748	0.6891

Number of obs: 98, groups: date, 43

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.54145	1.03697	45.43586	0.522	0.604105
sitecoal_2	-0.69845	0.68906	36.35111	-1.014	0.317458
sitecoal_3	-0.01475	0.68906	36.35111	-0.021	0.983039
sitecoal_4	-1.92835	0.68906	36.35111	-2.799	0.008163 **
sitecoal_5	-2.58865	0.68906	36.35111	-3.757	0.000603 ***
sitecoal_6	-1.90100	0.68906	36.35111	-2.759	0.009026 **
siteitalian_1	3.19845	1.79609	45.43586	1.781	0.081633 .
siteitalian_2	2.67715	1.79609	45.43586	1.491	0.142991
siteitalian_3	2.78385	1.79609	45.43586	1.550	0.128091
siteitalian_4	2.92835	1.79609	45.43586	1.630	0.109930
siteitalian_5	2.72605	1.79609	45.43586	1.518	0.136000
sitetc_1	-1.22304	1.20615	65.40299	-1.014	0.314317
sitetc_15	-1.65284	1.20615	65.40299	-1.370	0.175265
sitetc_2	-2.59424	1.20615	65.40299	-2.151	0.035189 *
sitetc_3	-3.44214	1.20615	65.40299	-2.854	0.005779 **
sitetc_ds	-0.44794	1.20615	65.40299	-0.371	0.711554
sitetc_isco	-2.78707	1.06395	45.62132	-2.620	0.011914 *
sitetc_mid	-0.80694	1.20615	65.40299	-0.669	0.505836
sitetc_us	-0.49374	1.20615	65.40299	-0.409	0.683618
siteutc_isco	-2.14744	1.06549	45.84336	-2.015	0.049742 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

fit warnings:

fixed-effect model matrix is rank deficient so dropping 60 columns / coefficients

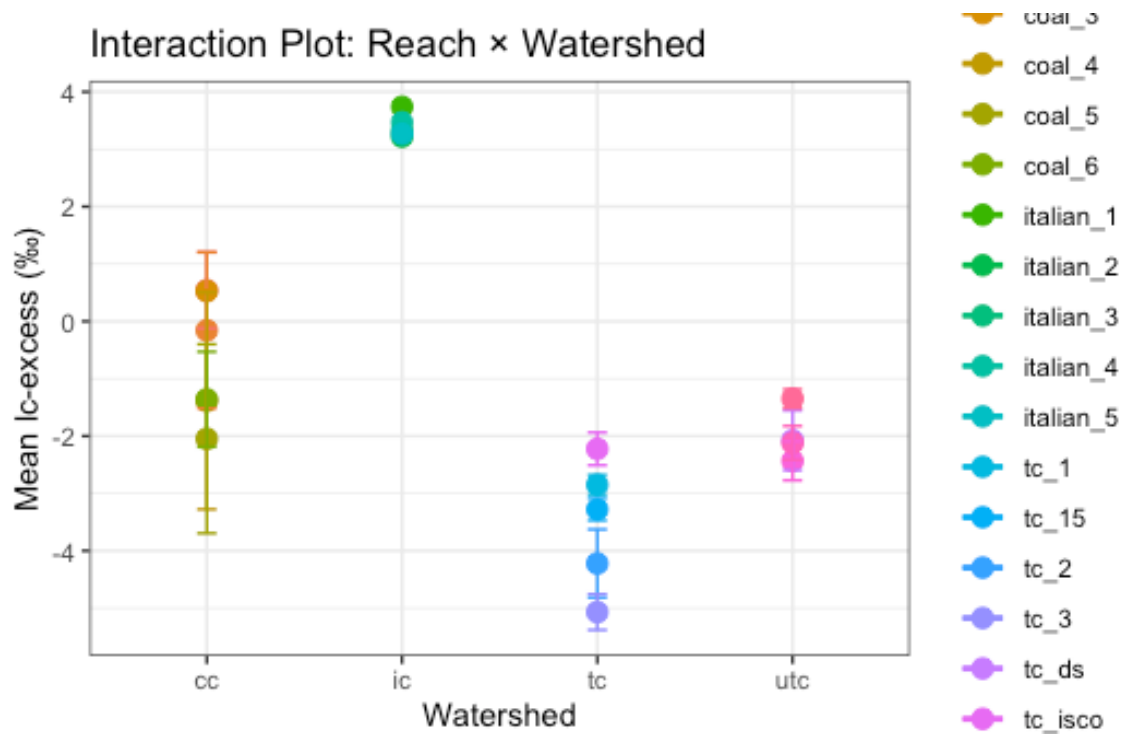
```
anova(model_mixed) # Type III tests via lmerTest
```

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
site	44.331	2.3332	19	38.53	4.9141	1.402e-05 ***
location						
site:location						

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
ggplot(compiled_isotope, aes(x = location, y = lc_excess, color = site, group = site)) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  stat_summary(fun = mean, geom = "line", linewidth = 1) +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1) +
  labs(title = "Interaction Plot: Reach \u00d7 Watershed",
       x = "Watershed", y = "Mean lc-excess (\u2030)", color = "Reach") +
  theme_bw()
```



```
delta_summary <- compiled_isotope |>
  group_by(site) |>
  summarize(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(
    aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
```

```

    width = 0.3,
    color = "black"
  ) +
  theme_minimal() +
  labs(
    title = "Mean LC Excess by Site with Standard Error",
    x = "Sampling Site",
    y = "Mean LC Excess"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Tilts labels so they
    don't overlap
    legend.position = "none"
  )

```



```

# Build a lookup table of stream distances, then compute reach boundaries
distances <- tibble(
  site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
           "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
  dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)
) %>%
mutate(

```

```

    xmax = dist_m,
    xmin = lag(dist_m, default = 0) # previous site's distance, 0 for first
  )

gradient_data_clean <- gradient_data %>%
  select(site, normalized_stream_distance_m,
         `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`) %>%
  pivot_longer(
    cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`),
    names_to = "date",
    values_to = "gradient"
  ) %>%
  mutate(
    date = case_when(
      date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
      date == "Δlc-excess/Δx_6-30" ~ "2026-06-30"
    ),
    gradient = as.numeric(gradient)
  ) %>%
  left_join(distances %>% select(site, xmin, xmax), by = "site") %>%
  mutate(
    site = factor(site, levels = distances$site)
  ) %>%
  filter(!is.na(gradient))

```

```

Warning: There was 1 warning in `mutate()`.
i In argument: `gradient = as.numeric(gradient)`.
Caused by warning:
! NAs introduced by coercion

```

```

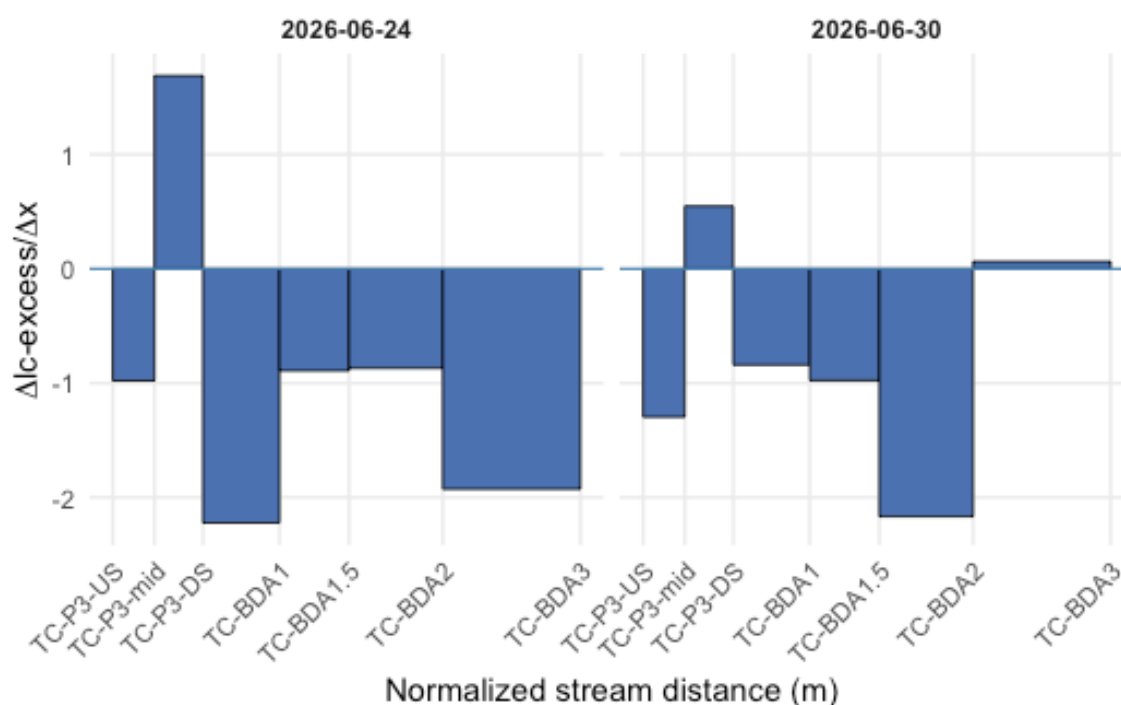
ggplot(gradient_data_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "#4C72B0", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "steelblue", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site
  ) +
  labs(
    x = "Normalized stream distance (m)",

```

```

y = expression(Delta*"lc-excess/"*Delta*"x")
) +
theme_minimal(base_size = 11) +
theme(
  panel.grid.minor = element_blank(),
  strip.text = element_text(face = "bold"),
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```



All reaches show lc-excess decreasing downstream, on both dates. Since lc-excess becoming more negative = more evaporative enrichment, this version says every single reach along Trail Creek is adding evaporative signal, with no recovery zones. That's a materially different story than "some reaches enrich, some reaches dilute" — it now reads as continuous, cumulative evaporative loss along the whole restored gradient.

Where the deepest enrichment is happening differs by date:

6-24: TC-P3-DS reach is the sharpest dip (~-2.2), essentially a local hotspot just downstream of the mid-P3 restoration point. 6-30: TC-BDA1.5→BDA2 is now the sharpest dip (~-2.15), while TC-P3-DS has relaxed to a much smaller magnitude (~-0.55) compared to 6-24.

TC-BDA3 is the outlier reach on both dates — it's close to flat (near-zero gradient), meaning that final reach adds almost no additional evaporative signal, regardless of date. That's consistent

between both panels and is probably your most robust finding here: whatever's happening upstream, the lowest reach isn't contributing much additional fractionation.

Exploratory Visuals of Coal Creek

Exploratory Visuals of Italian Creek